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public static int task6CP(int[] input) {
    int n = input.length;           // Length of the input array
    int maxLength = 0;              // Variable to store the max length
    int[] dp = new int[n];          // Array to store lengths of subsequences that end at each index.
    for (int i = 0; i < n; i++) {   // Loop through each element of the input array
        dp[i] = 1;                  // Initialize the length of the subsequence to 1
        for (int j = 0; j < i; j++) { // Loop through each element before the current element
            if (input[i] > input[j]) { // If the current element can be included in the increasing subsequence
                dp[i] = Math.max(dp[i], dp[j] + 1); // Update the length of the subsequence if necessary
            }
        }
    }

    for (int length : dp) {
        maxLength = Math.max(maxLength, length); // Find the maximum length from the array
    }

    return maxLength;               // Return the maximum length
}

public static void main(String[] args) {
    int[] input = {10, 9, 2, 5, 3, 7, 101, 18};
    int result = task6CP(input);
    System.out.println(result);
}

```